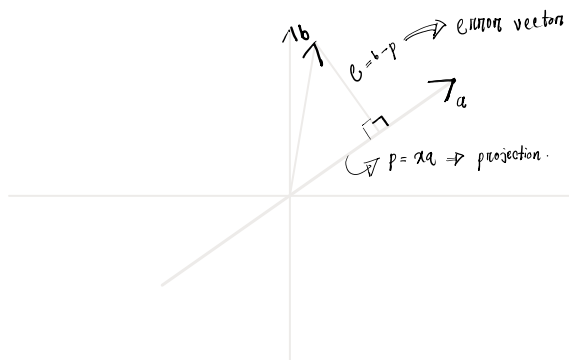


## Projection onto a line



Here,  $a$  is perpendicular to  $e$ .

$$\Rightarrow a \cdot e = 0$$

$$\Rightarrow a^T e = 0$$

$$\Rightarrow a^T (b - p) = 0$$

$$\Rightarrow a^T (b - xa) = 0$$

$$\Rightarrow xa^T a = a^T b$$

$$\Rightarrow x = \frac{a^T b}{a^T a} \quad | \quad p = xa$$

$$\therefore p = a \frac{a^T b}{a^T a} = \left( a \frac{a^T}{a^T a} \right) b$$

$$\hookrightarrow \text{proj } p = Pb$$

$$P = \frac{aa^T}{a^T a} \quad \begin{array}{l} \text{a full-rank} \\ \text{matrix} \end{array} \quad \begin{array}{l} \text{a number} \\ a^2 \end{array} \quad \Rightarrow \text{Projection matrix}$$

Now,  $C(P)$  = line through  $a$

$$\text{rank}(P) = 1$$

$$\textcircled{*} P^T = P \quad \text{as } P \text{ is symmetric}$$

Why project?

Because  $Ax=b$  may have no solution.

Instead solve  $A^T x = p$ ; proj of  $b$  onto column space

Example 1: Project  $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  onto  $a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$  to find  $p = xa$ .

Soln.

Here.

$$a^T b = (1+1) + (2+1) = 5$$

$$a^T a = (1+1) + (2+2) = 6$$

$$\text{where, } x = \frac{a^T b}{a^T a} = \frac{5}{6}$$

projection vector,  $p = a \hat{x} = a \frac{a^T b}{a^T a}$    
 $\hat{x}$  scalar value   
 $\frac{a^T b}{a^T a}$  length of  $a$   $\rightarrow$  scalar value   
 $\hookrightarrow \|a\|$    
 $\Rightarrow p = \left( \frac{a a^T}{a^T a} \right) b$   $\hookrightarrow$   $n \times n$  matrix   
 $\Rightarrow p = Pb$    
 $\hookrightarrow$  Projection Matrix   
 How am I separating a vector from scalar value?

$$\text{Now, } p = \hat{a} = \frac{5}{9}(1, 2, 2) = \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right)$$

Now, if we want to error vector  $e$ ,

$$\begin{aligned} e &= b - p \\ &= (1, 1, 1) - \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right) \\ &= \left(\frac{4}{9}, \frac{-1}{9}, \frac{-1}{9}\right) \end{aligned}$$

Now, if we want to confirm the correctness of our projection, the error vector  $e$  must be perpendicular to  $a$ .

$$\text{So, } e^T a = 0$$

$$\text{Now, } e^T a = \frac{4}{9} - \frac{1}{9} \times 2 - \frac{1}{9} \times 2 = 0$$

⊛ The vector  $b$  is split into two parts - its component along the line is  $p$ , its perpendicular part is  $e$ . Those two sides  $p$  and  $e$  have length  $\|p\| = \|b\| \cos \theta$  and  $\|e\| = \|b\| \sin \theta$ .

$$\text{Now, } p = \frac{a^T b}{a^T a} a \Rightarrow \|p\| = \frac{\|a\| \|b\| \cos \theta}{\|a\|^2} \|a\| = \|b\| \cos \theta$$

Example 2: Find the projection matrix  $P = \frac{aa^T}{a^T a}$  onto the line through  $a = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Sol<sup>n</sup>: Now,  $aa^T = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$

$$a^T a = 1^2 + 2^2 + 2^2 = 9$$

$$\text{Projection Matrix, } P = \frac{aa^T}{a^T a} = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$$

To project a vector  $b$  onto  $a$ , compute  $p = Pb$  for  $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$p = Pb = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5 \\ 10 \\ 10 \end{bmatrix}$$

$$\text{The error, } e = b - p = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 5 \\ 10 \\ 10 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 4 \\ -1 \\ -1 \end{bmatrix}$$

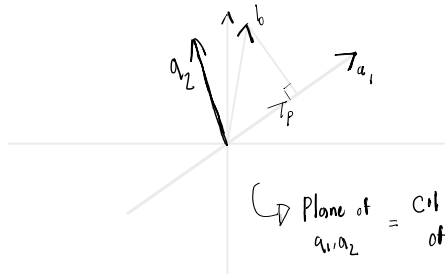
⊛ Projection matrix has idempotent property. Meaning,  $P^2 = P$  (will be proved later on)

⊛ The matrix  $I - P$  projects onto the orthogonal complement of  $a$ , which is the plane perpendicular to  $a$ .

$$(I - P)b = b - Pb = e$$

Thus,  $I - P$  produces the error vector  $e$ , which lies in the plane perpendicular to  $a$ .

## Projection Onto a Subspace



$$\hookrightarrow \text{Plane of } \begin{matrix} a_1, a_2 \\ \text{of } A \end{matrix} = \text{Col space} = \begin{bmatrix} 1 & 1 \\ a_1 & a_2 \end{bmatrix}$$

$$\hookrightarrow p = \hat{x}_1 a_1 + \hat{x}_2 a_2$$

$$\hookrightarrow p = A\hat{x}, \text{ so Find } \hat{x}$$

Key:  $b - A\hat{x}$  is perpendicular to plane

$$\text{So, } \begin{cases} a_1^T(b - A\hat{x}) = 0 \\ a_2^T(b - A\hat{x}) = 0 \end{cases} \rightarrow \begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix} (b - A\hat{x}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\hookrightarrow A^T(b - A\hat{x}) = 0 \quad \begin{matrix} \text{c} \\ \text{c} \end{matrix} \rightarrow \text{c in } N(A^T) \Rightarrow \text{c} \perp C(A)$$

The error vector  $b - A\hat{x}$  is perpendicular to the subspace.  
The error  $b - A\hat{x}$  makes a right angle with all vectors (here  $a_1, a_2$ ) in the subspace.

$$\text{So, } A^T A \hat{x} = A^T b$$

$$\Rightarrow \hat{x} = (A^T A)^{-1} A^T b$$

$$\therefore p = A\hat{x} = A(A^T A)^{-1} A^T b$$

$$\hookrightarrow \text{projection matrix } P = A(A^T A)^{-1} A^T$$

$$\text{Here, } P^T = P$$

$$P^2 = P$$

$$\text{Trace}(P) = n(P)$$

In  $\mathbb{R}^n$ ,  $n=1$ , the subspace is a line,

$$A^T A \Rightarrow a^T a$$

$$\therefore P = \frac{a a^T}{a^T a} \rightarrow \text{matches the formula for projection onto a line}$$

Example 3:  $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$  - find  $\hat{x}$ ,  $p$  and  $P$ .

Sol<sup>n</sup>:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

$$\text{Now, } A^T A \hat{x} = A^T b$$

$$\hookrightarrow \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}_{2 \times 2} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}_{2 \times 1} \Rightarrow \begin{cases} 3\hat{x}_1 + 3\hat{x}_2 = 6 \\ 3\hat{x}_1 + 5\hat{x}_2 = 0 \end{cases}$$

$$\hookrightarrow \hat{x}_1 = 5$$

$$\hookrightarrow \hat{x} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

$$x_2 = -3$$

Now, projection  $p = A\hat{x}$

$$= 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

$$\therefore e = b - p = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Now, projection matrix,  $P = A(A^T A)^{-1} A^T$

$$(A^T A)^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$$

$$\therefore A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

~~⊗~~ Proof that,  $P^2 = P$ .

Proof: We know that  $P = A(A^T A)^{-1} A^T$ .

$$\text{Now, } P^2 = P \cdot P$$

$$= A(A^T A)^{-1} A^T \cdot A(A^T A)^{-1} A^T$$

$$= A(A^T A^{-1}) \underbrace{A^T A (A^T A)^{-1}}_{\hookrightarrow I} A^T$$

$$= A(A^T A^{-1}) A^T$$

$$= P.$$

Some Key Points :

⊗ In the projection matrix  $p = A(A^T A)^{-1} A^T \rightarrow (A^T A)^{-1} \neq A^{-1}(A^T)^{-1}$

Because,  $A$  might not be a rectangular matrix. It has no inverse matrix.

⊗ When  $A$  is a  $n \times n$  matrix with linearly independent columns,  $A^T A$  is a square, symmetric and invertible matrix.

⊗ To find the projection  $p = \hat{x}_1 a_1 + \hat{x}_2 a_2 + \dots + \hat{x}_n a_n$ , solve the normal equation:

$$A^T A \hat{x} = A^T b.$$

⊗ The projection is:  $p = A \hat{x}$

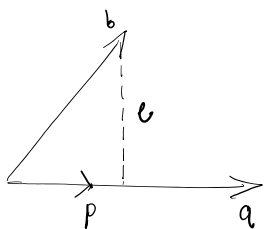
⊗ The error is:  $e = b - p = b - A \hat{x}$

⊗ The projection matrix is:  $P = A(A^T A)^{-1} A^T$

⊗ The distance from  $b$  to the subspace  $C(A)$  is  $\|e\|$ .

⊗ When  $P$  projects onto one subspace,  $I - P$  projects onto the perpendicular subspace.

How so?



$$\hookrightarrow Pb = p$$

We can write,

$$b = p + e$$

$$\Rightarrow b = Pb + e$$

$$\Rightarrow b - Pb = e.$$

$$\Rightarrow (I - P)b = e.$$

⊗ Why the  $\text{trace}(P) = \text{rank}(P)$ ?

$\hookrightarrow$  Will be explained in C-1

Example 1: Find the projection matrix onto the plane:  $x + y - z = 0$

Sol<sup>n</sup>: plane  $\Rightarrow x + y - z = 0$

$$\hookrightarrow \begin{bmatrix} 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Taking  $(1, 1, -2)$  and  $(1, 0, 1)$  as basis vector.

$$\text{So, } A = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$\therefore P = A(A^T A)^{-1} A^T$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & -2 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 5 & -9 \\ 5 & 6 & -13 \\ -9 & -13 & 20 \end{bmatrix}$$

$$\left| \begin{array}{l} A^T A = \begin{bmatrix} 6 & -1 \\ -1 & 2 \end{bmatrix} \\ (A^T A)^{-1} = \frac{1}{11} \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix} \end{array} \right.$$

Example 2: Find the projection matrix that projects any vector on the plane  $x - y - 2z = 0$

Sol<sup>n</sup>: plane  $\Rightarrow x - y - 2z = 0$

$$\hookrightarrow \begin{bmatrix} 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Taking  $(1, 1, 0)$  and  $(2, 0, 1)$  as basis vectors.

$$\text{So, } A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore P = A(A^T A)^{-1} A^T$$

$$= \frac{1}{6} \begin{bmatrix} 5 & 1 & 1 \\ 1 & 5 & -2 \\ 2 & -2 & 2 \end{bmatrix}$$